

# Homework 1 Group I

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## Question 1

Let  $s_i = 1 - 2y_i$ . For  $\lambda > 0$ , the objective function is

$$f_\lambda(\theta) = \underbrace{\sum_{i=1}^m \phi(s_i \langle \tilde{x}_i, \theta \rangle)}_{f(\theta)} + \frac{\lambda}{2} \|\theta\|^2, \quad \theta \in E.$$

We first show that  $f$  is convex. The smoothed hinge loss  $\phi$  satisfies

$$\phi'(z) = \begin{cases} 0, & z \leq -1, \\ 1 + z, & -1 < z < 0, \\ 1, & z \geq 0. \end{cases}$$

The values and piecewise derivatives agree at  $z = -1$  and  $z = 0$ . Thus  $\phi$  is continuously differentiable. Its derivative is nondecreasing, so for any  $a, b \in \mathbb{R}$ ,

$$\phi(b) - \phi(a) - \phi'(a)(b - a) = \int_a^b (\phi'(u) - \phi'(a)) du \geq 0.$$

(The inequality holds for either ordering of  $a$  and  $b$ .) By Theorem 4.21 of the lecture notes,  $\phi$  is convex.

For any  $\theta, \eta \in E$  and  $t \in [0, 1]$ , linearity of the inner product and convexity of  $\phi$  give

$$\begin{aligned} \phi(s_i \langle \tilde{x}_i, (1-t)\theta + t\eta \rangle) &= \phi((1-t)s_i \langle \tilde{x}_i, \theta \rangle + ts_i \langle \tilde{x}_i, \eta \rangle) \\ &\leq (1-t)\phi(s_i \langle \tilde{x}_i, \theta \rangle) + t\phi(s_i \langle \tilde{x}_i, \eta \rangle). \end{aligned}$$

Summing over  $i$  yields  $f((1-t)\theta + t\eta) \leq (1-t)f(\theta) + tf(\eta)$ ; hence  $f$  is convex.

Now

$$f_\lambda(\theta) - \frac{\lambda}{2} \|\theta\|^2 = f(\theta)$$

is convex. Since  $\lambda > 0$ , Definition 4.5 of the lecture notes therefore implies that  $f_\lambda$  is  $\lambda$ -strongly convex. It is real valued on the whole Euclidean space  $E$ , so Theorem 4.28 applies:  $f_\lambda$  has a unique global minimizer  $\theta^*$ .

Moreover,  $f_\lambda$  is convex and differentiable on  $E$ . Each loss term is a composition of the continuously differentiable function  $\phi$  with a linear map, and the quadratic term is differentiable. By Corollary 4.23 of the lecture notes, a point is a global minimizer if and only if its gradient vanishes. So, what we can say about the critical points is that

$$\{\theta \in E : \nabla f_\lambda(\theta) = 0\} = \{\theta^*\}.$$

## Question 2

We now set  $\lambda = 0.005$ . Write  $g_i(\theta) = s_i \langle \tilde{x}_i, \theta \rangle$ . Each  $g_i : E \rightarrow \mathbb{R}$  is linear, hence differentiable, and  $\phi$  is continuously differentiable as shown in Question 1. Thus  $f = \sum_{i=1}^m \phi \circ g_i$  is differentiable. By additivity of the differential and the chain rule, for any  $h \in E$ ,

$$Df(\theta)[h] = \sum_{i=1}^m \phi'(g_i(\theta)) Dg_i(\theta)[h].$$

Using the limit expression for the differential of  $g_i$ , we obtain

$$Dg_i(\theta)[h] = \lim_{t \rightarrow 0} \frac{g_i(\theta + th) - g_i(\theta)}{t} = \lim_{t \rightarrow 0} \frac{ts_i \langle \tilde{x}_i, h \rangle}{t} = s_i \langle \tilde{x}_i, h \rangle.$$

Substituting and using bilinearity of the inner product gives

$$\begin{aligned} Df(\theta)[h] &= \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \langle \tilde{x}_i, h \rangle \\ &= \left\langle \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \tilde{x}_i, h \right\rangle. \end{aligned}$$

The quadratic term  $r(\theta) = \frac{\lambda}{2} \|\theta\|^2$  is also differentiable. Indeed,

$$r(\theta + h) - r(\theta) - \lambda \langle \theta, h \rangle = \frac{\lambda}{2} \|h\|^2 = o(\|h\|),$$

so  $Dr(\theta)[h] = \lambda \langle \theta, h \rangle$ . Consequently,  $f_\lambda = f + r$  is differentiable and

$$Df_\lambda(\theta)[h] = \left\langle \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \tilde{x}_i + \lambda \theta, h \right\rangle.$$

By Definition 2.18 of the lecture notes,  $Df_\lambda(\theta)[h] = \langle \nabla f_\lambda(\theta), h \rangle$ . Since this equality holds for every  $h \in E$ , we conclude that

$$\nabla f_\lambda(\theta) = \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \tilde{x}_i + \lambda \theta.$$

**Numerical agreement and runtime.** We checked the objective values and each gradient component using `np.testing.assert_allclose`, with absolute and relative tolerances of  $10^{-10}$ . Both implementations agreed within these tolerances. We performed 30 timing trials on identical inputs using `timeit.repeat`, retaining the minimum of three single evaluations for each implementation in each trial. Across these trials, the lowest observed speedups from vectorization were approximately  $15.90\times$  for the objective and  $10.30\times$  for the gradient.

### Question 3

To check our implementation of the gradient, we use a first-order Taylor expansion of  $f_\lambda$ . We choose a random point  $\theta \in \mathbb{R}^{785}$  and a random unit vector  $v \in \mathbb{R}^{785}$ , and consider

$$E(t) = |f_\lambda(\theta + tv) - f_\lambda(\theta) - t \langle \nabla f_\lambda(\theta), v \rangle|.$$

If the implemented gradient is correct, then the first-order Taylor expansion predicts

$$f_\lambda(\theta + tv) = f_\lambda(\theta) + t \langle \nabla f_\lambda(\theta), v \rangle + O(t^2),$$

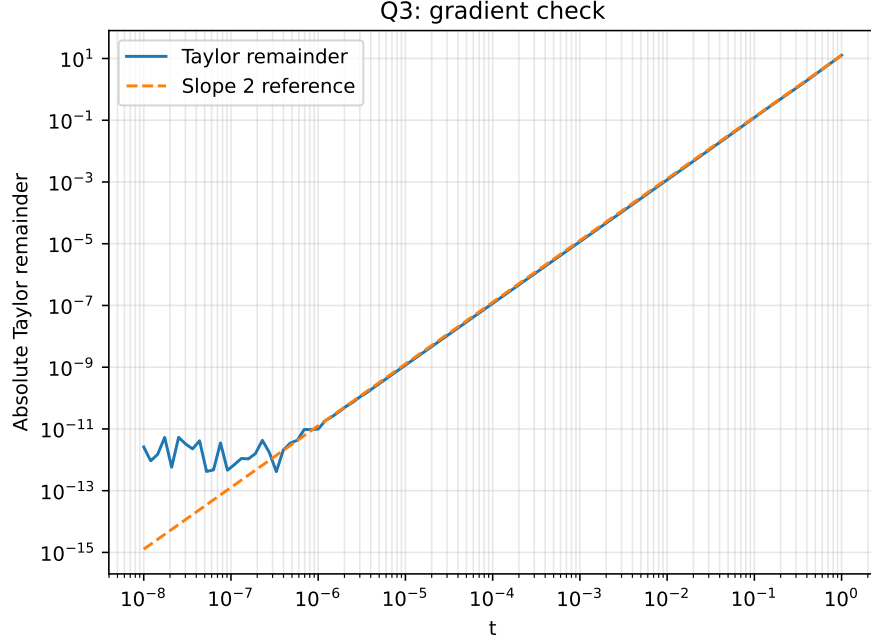
and therefore

$$E(t) = O(t^2).$$

We evaluate  $E(t)$  for

$$t \in \text{logspace}(-8, 0, 101)$$

and plot the result on logarithmic axes.



On the approximately straight part of the log–log plot, the observed slope is close to 2. Indeed, if

$$E(t) \approx Ct^2,$$

then

$$\log E(t) \approx \log C + 2 \log t,$$

so the expected slope is 2.

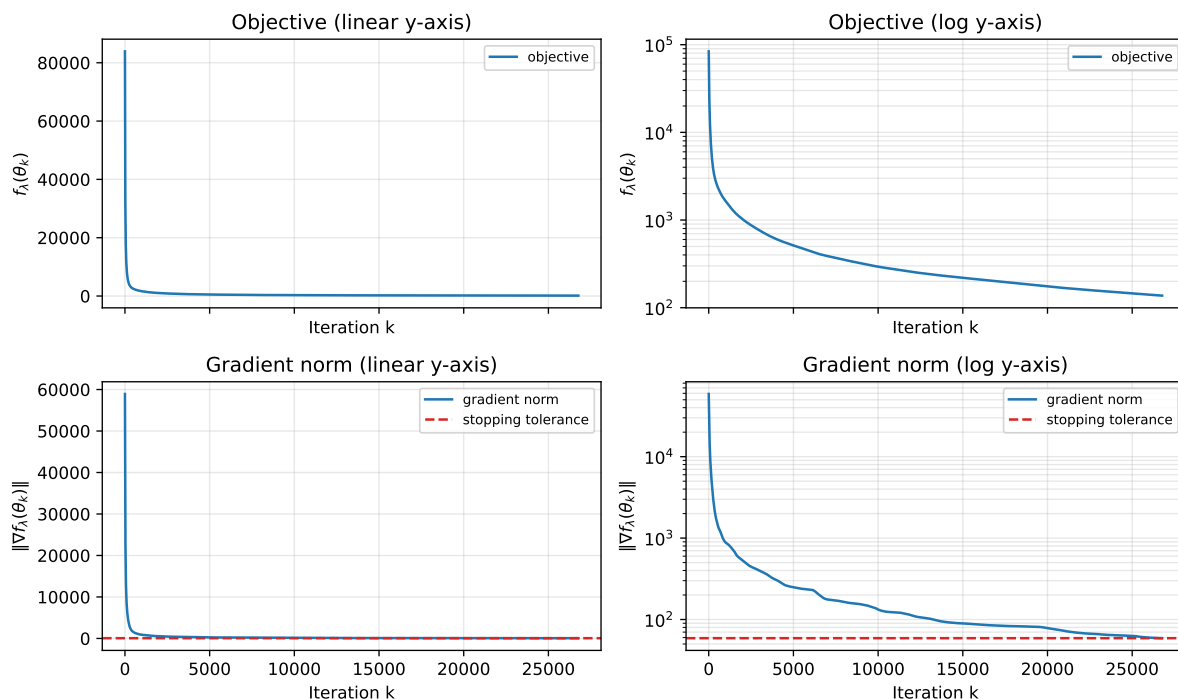
For very small values of  $t$ , the error no longer follows the quadratic behavior and becomes irregular. This is due to floating-point roundoff and cancellation when subtracting nearly equal quantities.

Thus, the observed slope is consistent with the expected second-order Taylor remainder and provides numerical evidence that our gradient implementation is correct. However, this does not constitute a proof of correctness, since the test is performed only at a particular point and direction and uses finite-precision arithmetic.

## Question 4

Given that  $\nabla f_\lambda$  is  $L$ -Lipschitz continuous with constant  $L = \sigma_{\max}(X)^2 + \lambda$  (where  $\sigma_{\max}(X)$  is computed numerically), we select the constant step-size  $\alpha = \frac{1}{L} = \frac{1}{\sigma_{\max}(X)^2 + \lambda}$  as a consequence of Theorems 3.3 and 3.4. Moreover, since  $f_\lambda$  is  $\lambda$ -strongly convex, we know by Theorem 4.31 and Corollary 4.32 that  $f_\lambda(\theta_k)$  (respectively  $\|\nabla f_\lambda(\theta_k)\|$ ) converges at least linearly to the unique global minimum  $f_\lambda(\theta^*)$  (respectively 0).

## Question 5



The four graphs compare the objective value and gradient norm against iteration number using linear (left) and logarithmic (right) vertical axes. We prefer logarithmic vertical axes because both quantities decrease over several orders of magnitude, and the linear scale hides much of the later progress. A geometric bound  $e_k \leq Cq^k$ , with  $0 < q < 1$ , becomes a linear upper bound on  $\log e_k$ . Roughly straight descending portions of a semilog curve can therefore suggest linear convergence, but do not establish such a bound. The objective panel shows  $f_\lambda(\theta_k)$ , not the gap  $f_\lambda(\theta_k) - f_\lambda(\theta^*)$ ; its flattening is not evidence against linear convergence. The theoretical guarantee is discussed in Question 6.

## Question 6

Consider the fixed-step gradient descent iteration

$$\theta_{k+1} = \theta_k - \alpha \nabla f_\lambda(\theta_k), \quad k \geq 0.$$

From Question 1,  $f_\lambda$  is  $\lambda$ -strongly convex since

$$f_\lambda(\theta) - \frac{\lambda}{2} \|\theta\|^2 = f(\theta)$$

is convex. Moreover, as stated in the homework,

$$\nabla f_\lambda$$

is Lipschitz continuous with constant

$$L = \sigma_{\max}(X)^2 + \lambda.$$

Hence the assumptions of Theorem 4.31 are satisfied with

$$\mu = \lambda, \quad L = \sigma_{\max}(X)^2 + \lambda.$$

Therefore  $f_\lambda$  has a unique minimizer  $\theta^\star$ , and for the step size

$$\alpha = \frac{1}{L} = \frac{1}{\sigma_{\max}(X)^2 + \lambda},$$

Theorem 4.31 gives

$$f_\lambda(\theta_k) - f_\lambda(\theta^\star) \leq \left(1 - \frac{\lambda}{L}\right)^k (f_\lambda(\theta_0) - f_\lambda(\theta^\star)).$$

Thus the objective values converge globally and at least linearly to the optimum.

More generally, for every

$$0 < \alpha \leq \frac{1}{L},$$

the same theorem may be applied with Lipschitz constant  $1/\alpha$ , yielding

$$f_\lambda(\theta_k) - f_\lambda(\theta^\star) \leq (1 - \alpha\lambda)^k (f_\lambda(\theta_0) - f_\lambda(\theta^\star)).$$

For the value  $\lambda = 0.005$ , the admissible step sizes are therefore

$$0 < \alpha \leq \frac{1}{\sigma_{\max}(X)^2 + 0.005}.$$

## Application to our implementation

Our implementation uses exactly

$$\alpha = \frac{1}{L}, \quad L = \sigma_{\max}(X)^2 + \lambda,$$

so Theorem 4.31 applies directly.

Writing

$$\kappa = \frac{L}{\lambda},$$

we obtain

$$f_\lambda(\theta_k) - f_\lambda(\theta^\star) \leq \left(1 - \frac{1}{\kappa}\right)^k (f_\lambda(\theta_0) - f_\lambda(\theta^\star)).$$

Corollary 4.32 further gives

$$\|\theta_k - \theta^\star\| \leq \sqrt{\frac{2}{\lambda} (f_\lambda(\theta_0) - f_\lambda(\theta^\star))} \left(\sqrt{1 - \frac{1}{\kappa}}\right)^k$$

and

$$\|\nabla f_\lambda(\theta_k)\| \leq \sqrt{2L (f_\lambda(\theta_0) - f_\lambda(\theta^\star))} \left(\sqrt{1 - \frac{1}{\kappa}}\right)^k.$$

Hence both the iterates and the gradient norm converge at least linearly.

## Comparison with the finite run

This prediction is consistent with our numerical experiment. The plot shows  $\|\nabla f_\lambda(\theta_k)\|$  decreasing rapidly at first and then more slowly, until the stopping tolerance is reached after approximately  $2.7 \times 10^4$  iterations.

Corollary 4.32 only provides the geometric upper bound

$$\|\nabla f_\lambda(\theta_k)\| \leq Cq^k, \quad q = \sqrt{1 - \frac{\lambda}{L}} < 1,$$

rather than requiring the gradient norm to decrease by the same factor at every iteration. Thus the changes of slope and relatively flat regions in the plot are compatible with the theory.

Since the numerical run stops once the prescribed tolerance is reached, it does not show convergence all the way to zero, but its behavior is consistent with the asymptotic prediction

$$\|\nabla f_\lambda(\theta_k)\| \rightarrow 0.$$

## Question 7

The classification error is approximately 0.134% on the training set and 0.189% on the test set. The test error is slightly higher than the training error, as expected, but remains very small, indicating that the classifier generalizes well to unseen samples.

### Q8. Mathematical insight

The regularization term

$$\frac{\lambda}{2}\|w\|^2$$

makes  $f_\lambda$   $\lambda$ -strongly convex. Therefore,  $f_\lambda$  has a unique minimizer  $w^*$ , and gradient descent converges to  $w^*$  from any starting point. In particular, with the step size

$$\alpha = \frac{1}{L},$$

where  $L$  is the Lipschitz constant of  $\nabla f_\lambda$ , gradient descent converges at least linearly.

### Computational insight

Vectorizing the computations significantly improves the efficiency of the implementation. Instead of evaluating the objective and gradient with explicit Python loops, we use matrix operations such as

$$X^\top(X\theta - y).$$

These operations are executed by optimized numerical linear algebra routines and are therefore much faster.

In our experiments, across 30 trials, the median speedup was approximately 18.91 times for the objective and 12.62 for the gradient. Since gradient descent requires many iterations, this speed-up has a substantial impact on the total runtime.